

Performance Testing

Introduction

Performance Testing is conducted to evaluate the responsiveness, stability, and efficiency of the Metro Ticket Generation System under different operating conditions. The objective is to ensure that the application performs reliably with multiple users and processes ticket bookings without delays.

Objectives

- Measure system response time.
- Verify fare calculation performance.
- Ensure QR ticket generation is completed quickly.
- Evaluate system behavior during multiple booking requests.
- Confirm smooth operation without performance degradation.

Performance Test Scenarios

Test ID	Test Scenario	Expected Result	Status
PT-01	Open Metro Ticket Catalog	Catalog loads within 2 seconds	Pass
PT-02	Load Metro Station List	Stations displayed correctly	Pass
PT-03	Submit Booking Request	Booking submitted successfully	Pass
PT-04	Calculate Fare	Fare displayed instantly	Pass
PT-05	Generate Ticket Number	Ticket generated automatically	Pass
PT-06	Generate QR Ticket	QR generated within 2 seconds	Pass
PT-07	Store Booking Record	Record saved successfully	Pass
PT-08	Email Notification	Email received within 1 minute	Pass
PT-09	Multiple Concurrent Bookings	System handles requests without failure	Pass
PT-10	View Booking History	Records load successfully	Pass

Performance Metrics

Parameter	Expected Value	Actual Result
Catalog Loading Time	≤ 2 sec	1.5 sec
Fare Calculation	≤ 1 sec	0.8 sec
Ticket Generation	≤ 2 sec	1.7 sec
QR Generation	≤ 2 sec	1.9 sec
Database Update	≤ 1 sec	0.6 sec
Email Notification	≤ 60 sec	35 sec

Performance Test Summary

The Metro Ticket Generation System demonstrated satisfactory performance during testing. All booking operations, fare calculations, QR generation, and notification processes completed within the expected response times. The application remained stable during concurrent booking requests and maintained accurate data processing throughout the testing phase.